**TITLE AND DATE**

Document name: *Project#1 Asynchronous Tasking*
Document reference: *Poulos, Theo*.cmpe311.fall26.*project#1,Project#2 Using a Cyclic Executive Manager*
Date of publication: 3/13/26

**LEAD ENGINEER:** *Theo Poulos, CMPE 311 Student*

**STAKEHOLDERS:**
Prof Kidd, Instructor, University of Maryland Baltimore County, Baltimore, Maryland, USA
MD Safwan Zaman, TA, University of Maryland Baltimore County, Baltimore, Maryland, USA

**HIGH-LEVEL DESCRIPTION:**
This project will use C programming language to program an arduino where we may use the arduino.h library. We will be reading input from the user to determine which LED should turn on and for what interval. The data will then be sent through the serial ports to a Breadboard with 2 LEDS connected with 2 200Ω resistors.

**DESCRIPTION:**
This project is designed to use an embedded system where, over the serial port, you ask the user which LED and at what interval they want to turn on. The arduino will then have the LED specified blink at the rate defined independently of each other. Each LED should be able to be at any rate and be on or off independently.

**DEFINITIONS AND ABBREVIATIONS:**
Arduino – an Italian open-source hardware and software company; also refers to a development board created by that company
arduino.h – header for a library of convenience functions specific to the Arduino development platform
AVR – A family of microcontrollers, originally developed by Atmel, and currently owned by Microchip Technology
avr.h – header including helper functions and definitions for using an AVR microcontroller
ELEGOO – A Chinese company that develops and markets 3D printers and accessories
HL – High-level
IDE – Integrated Development Environment
gcc – front end for the GNU Compiler Collection
Github – A widely used distributed SVC (Software Version Control) system
LED – Light Emitting Diode

The Customer-requirements" – The requirements defined by The Customer as satisfying The Contract.

“The HL-requirements” – The System’s high-level technical requirements derived from The Customer-requirements.
“The Requirements” – context dependent; e.g. may refer to the HL-requirements or Customer-requirements
“The Educational-constraints” – The requirements imposed by the instructor unrelated to the embedded system that allows The System to be evaluated.
“The Company” – The organization The Customer has contracted with to build The System.
“The Contract” – The business document that legally binds The Company to provide some service or product to The Customer.
“serial-monitor” – The serial port used by the Arduino IDE to communicate with The User.
“The Reference-platform” – The configuration of The System used by The Developer to test and validate The System. For this class, The System is the Arduino compatible ELEGOO Uno R3 development board.

**CONVENTIONS:**
Must, shall or will – your design must satisfy the requirement
May – your design may satisfy the requirement but doesn’t have to
Informative – the intent of the following description is to make the requirement more understandable
All customer requirements are started with “C.#”.
All high-level requirements are started with “HL.#”.
All testing/validation requirements are started with “T.#”

**CUSTOMER REQUIREMENTS:**

The customer requirements, C1 through C5, are identical to that of PROJECT-ASYNC.

Ref: TheoPoulos.cmpe311.spring26.project#1, CUSTOMER REQUIREMENTS section.

**HIGH-LEVEL TECHNICAL REQUIREMENTS:**

The high-level technical requirements, HL.1 through HL.3, are identical to that of PROJECT-ASYNC.

Ref: TheoPoulos.cmpe311.spring26.project#1, HIGH-LEVEL TECHNICAL REQUIREMENTS section.

**TESTING AND VALIDATION REQUIREMENTS:**

The testing and validation requirements, T.1 through T.3, are identical to that of PROJECT-ASYNC.

Ref: TheoPoulos.cmpe311.spring26.project#1, HIGH-LEVEL TECHNICAL REQUIREMENTS section.

**DESIGN:**

The Design of the circuit is identical to that of the PROJECT-ASYNC.

Ref: TheoPoulos.cmpe311.spring26.project#1, HIGH-LEVEL TECHNICAL REQUIREMENTS section.

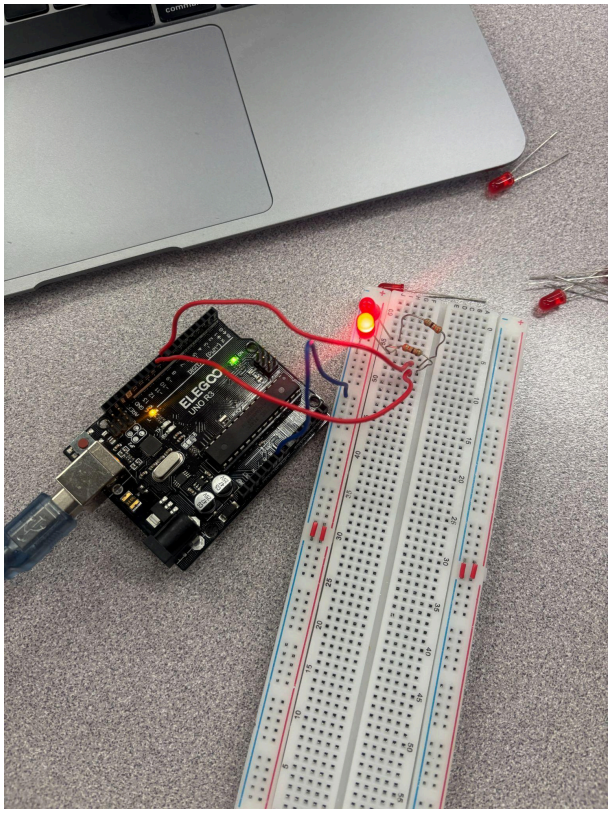
**TESTING:**

IMG_3505.mov

This video shows the testing of the system with the changes implemented. In the video you can see that the test matches the expected output

**ENGINEERING CHANGES:**

Changes consist of updated code switching from a standard loop to a round-robin cyclic executive task manager. The file that will be affected is the lights.ino code for the arduino. The change was implemented to make the code more efficient.

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